

VLSI DESIGN AND VERIFICATION

S.No.	Day Order	Theory	Lab
1	Day-1	Introduction to VLSI: * VLSI Design Flow * Types of Logic device * Advanced Technologies * Various fields and scope of VLSI Introduction to Digital Electronics * Digital Value set * Number Systems * Sign Magnitude, 1's, 2's, 9's and 10's Complement * SOP, POS * Boolean Algebra, K-map * Value set Conversion * Minimization Techniques and Theorem	Practical Exercise on Number Systems, SOP, POS and Simplification of Boolean expression
2	Day-2	* Combinational Circuit * Sequential Circuit * Different Types of FF * Synchronous and Asynchronous Counter * Memory Configuration and Conversions	Practical Exercise on building combinational and Sequential Logic
3	Day-3	BASICS OF VERILOG HDL : Importance of HDL - Design Methodologies - Basic Concepts - Data Types - Operators - Different types of Modeling - Dataflow Modeling - Gate Level Modeling - Behavioural Modeling: if, else, case statement.	Exercise on the Different Styles of Modelling. (Gate, Dataflow, Behavioural)
4	Day-4	ASSIGNMENTS AND BLOCKS OF STATEMENTS : Blocking and Non-blocking - Continuous Assignments - Assign And Deassign - Force And Release Delays - Inter and Intra Assignment Delay assignment, Blocks of Statement - Sequential Block - Parallel Block , Structured - Initial - Always.	Exercise on Blocking and Non-Blocking Assignments, Sequential Blocks

5	Day-5	ADVANCED VERILOG HDL AND TEST BENCHES : Parameter - Tasks - Functions - Timing and Delays - Verilog Test Benches for Combinational Logic Modules and Sequential Digital Circuits.	Exercise on Designing Fuctionality with Timing consuming and Non-Timing.
6	Day-6	DIGITAL DESIGN USING VERILOG HDL : Finite State Machine - FSM (Pattern Detector),Design ALU with Multiple Module,	Designing AMBA - APB protocol design.
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7	Day-7	BASICS OF VERIFICATION CONSTRUCTS & DATA TYPES Importance of Verification language and System Verilog(SV) – Difference between Verilog and SV – SV Data types – 2 State Data types (bit, byte, int, long_int, short_int) – 4 State Data types (integer, real, real-time, logic, reg) – User-defined data types (Struct, enum, string, typedef).	Exercise on Implementing Two state and four state Datatypes, User defined datatypes
8	Day-8	ARRAYS : SV Arrays – Pack & Unpack Arrays – Static Array – Dynamic Arrays (Dynamic, Associative and Queue Array), Structure and Union, Loops - while, do while, forever, for, foreach, repeat, break and continue.	Exercise on Implementing Array concepts and Managing with Built-in methods.
9	Day-9	Function and Task, System verilog Process Block, Class Constructor, Handle and Object Creation, This and super construct, class copy methods, Shallow and Deep copy methods	Exercise on Implementing Class with different methods to manage properties to assign and copy.
10	Day-10	OBJECT-ORIENTED PROGRAMMING AND METHODS : OOPs – Encapsulation – Polymorphism – Inheritance – Local – Protected – Public – Built-in Methods: Randomization, Pre and Post randomization, Constraint randomization and Types.	Exercise on Constructing class with oops concepts - data hiding technique, Creating sub-class and randomization of them.
11	Day-11	Scope resolution operator, Extern Methods, Constant Class properties, typedef class, Interprocess Synchronization Mechanism - Mailbox, Semaphore, Event.	Exercise on creating a class with Extern method, Communicating between SV Components to transfer packets through mailbox.
12	Day-12	Program Block, Scheduling Schematic, System verilog Interface Mechanism - Bundel , Directional information, Clocking Information, Virtual Interface,	Implementing Simple design using Interface to handle the static and dynamic environment
13	Day-13	Coverage - Functional and Code coverage, coverage, Coverpoint, cross coverage, bins	Verification using coverage and generate report
14	Day-14	Assertion-Immediate and Concurrent Assertion	Verification using Assertion
15	Day-15	Demo Project - SV TB Architecture for Combinational and sequential Circuit.	Assessment

**** Note : Open source tool is used for the training (If any related tool is available in the premises, can go with it).**